DCF Analysis

Chapter 9

Introduction to Financial Management (Pope)

Discounted Cash Flow (DCF) Analysis

DCF Analysis is a broad, inclusive term largely for:

- Preparation of cash flow estimates of a project
- Use of those estimates in capital budgeting analysis
- DCF is a part of the NPV calculation

Cash Flow Not Profit

Internal investment projects are attractive due to the cash flow they generate

- A firm can spend its operating cash flow but not its net
- income
- Accounting income (net income) includes many accrued and deferred items and may lead to incorrect decisions

$$FCF = EBIT(1 - T) + Depreciation - \Delta NWC - \Delta Gross\ PPE$$

Example 1:

A project costs $2,000 and is expected to last 2 years, producing cash income of $1,500 and $500 respectively. The cost of the project can be depreciated at $1,000 per year. Given a 10% required return, compare the NPV using cash flow to the NPV using accounting income.

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	C_0	C_1	C_2	NPV @ 10%
Cash Inflow		1500	500	
Project Cost	–2000			
Free Cash Flow	**-2000**	**1500**	**500**	**-233.14**
Depreciation		-1000	-1000	
Accounting (Net) Income		**500**	**-500**	**41.32**

Incremental Cash Flow

When conducting DCF analysis, you must discount the incremental free cash flows of a project

- Find the difference between the expected cash flows with a project minus the cash flows without the project
- Accounts for only the incremental (marginal) cash flows from an investment and not ones that would already occur
- Best to think about them in categories

Incremental Cash Flow

1. Include all indirect effects
   - Trace out any indirect impact on future cash flows
   - Cannibalization/Erosion costs: costs that arise when a new product or service competes with revenue generated by a current product or service
   - Synergy gains: increased revenues for other existing products related to the introduction of a new product

Incremental Cash Flow

2. Ignore sunk costs
   - Expenses that have already been incurred, or that will be incurred, regardless of the decision to accept or reject a project
   - Should not be considered as incremental cash flow
3. Include opportunity cost
   - Benefit or cash flow forgone as a result of an action
   - Compare “with or without” not “before versus after”
   - Examples: idle land or equipment
   - Should be included as incremental cash flow

Incremental Cash Flow

4. Recognize the investment in working capital
   - Net working capital is current assets minus current liabilities
   - Investment in current assets: inventories and accounts receivables
   - Investment in current liabilities: accounts payable and other accruals
   - ***Note: investments in working capital can change throughout a projects life and are recovered at the end of the project***

Incremental Cash Flow

5. Remember terminal cash flows
   - Recovery of investments in working capital
   - Disposal costs (both positive and negative)
6. Beware of allocated overhead costs
   - Costs that cannot conveniently be charged to any specific cost center (project) but are associated with running the business
   - Accounts allocate these to projects based on overhead rates

Investment vs Financing

Typically, you ignore how a project is financed when valuing it as an investment

- Separating investing and financing decisions
- Treat as all equity financed with all cash inflows going to shareholders (interest payments on debt not included in incremental cash flow)
- Ask the question: ***Is this project's existence dependent on financing?***

Forecasting Cash Flows

$$FCF = EBIT(1 - T) + Depreciation - \Delta NWC - \Delta Gross\ PPE$$

This is not meant to be an equation but rather a guide to think about measuring and including cash flows in a DCF analysis

Operating Cash Flow

OCF is what we generally care about when forecasting incremental cash flows

- **Method 1: Dollars in, dollars out:**

  OCF = revenues – expenses – taxes

- **Method 2: Adjusted accounting profit**

  OCF = NOPAT + depreciation

- **Method 3: Depreciation tax shield**

  OCF = (revenues – cash expenses) x (1 – tax rate) + (tax rate x depreciation)

Example 2:

Blooper Industries is looking to invest in a mining project that forecasting revenues of $15,000 and expenses of $10,000 in year 1. If the company depreciates their capital expenditure by $3,000 per year and has a marginal tax rate of 21%, what is the incremental cash flow from operations in year 1?

Example 2:

Blooper Industries is looking to invest in a mining project that forecasting revenues of $15,000 and expenses of $10,000 in year 1. If the company depreciates their capital expenditure by $3,000 per year and has a marginal tax rate of 21%, what is the incremental cash flow from operations in year 1?

Revenues	Revenues	$15,000
− Expenses	− COGS	10,000
− Depreciation	− Depreciation	3,000
= Pretax profit	= EBIT or EBT	$2,000
− Tax @ 21%	− Tax @ 21%	420
= NOPAT/Net Income	= EBIT - Tax	$1,580
+ Depreciation	+ Depreciation	3,000
= CF from operations	= CF from operations	$4,580

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Capital Investment

Capital expenditure generally involves an initial cash outflow, but attention should also be paid to the disposal of the capital equipment

- Disposal cost could be a cash outflow (true costs, cleaning, etc) or a cash inflow (salvage value)
- ***After-tax Salvage Value = Selling Price – Tax rate * (Selling price – Book Value)***

Example 1:

The manager of the Grand Junction Furniture Company decides to dispose of the specialty tools, acquired 2 years ago at a cost of $430,000 (including installation), to another firm for $125,000. How much of an after-tax cash flow will result, assuming that the tools were being depreciated based on a 3-year straight-line depreciation and the company's marginal tax rate is 35%?

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Depreciable basis = $430,000
Year 1 depreciation rate = 33.33%
Year 2 depreciation rate = 33.33%
Accumulated depreciation = $430,000 * (33.33% + 33.33%) = $286,638

Book Value = Depreciable basis – Accumulated depreciation
Book Value = $430,000 - $286,638 = $143,362
After-tax Salvage Value= Selling Price – Tax rate * (Selling price – Book Value)
=$125,000 - .35*($125,000-$143,362)
=$131,427

Depreciation

There are several ways a firm can handle depreciation:

1. **Straight-line Depreciation:**
   - Fixed depreciation expense over the life of an asset
   - Typically equals cost divided usable life
2. **Double-declining / MACRS**
   - Depreciation expense declines over life of asset
   - Typically straight-line expense x 2 per year
3. **“Bonus” depreciation**
   - Allows for 100% depreciation expense i

Example 3:

Blooper Industries is looking to invest in a mining project with a capital investment of $150,000 and cash flows given in the table below. Compare how the company's operating cash flows differ under a straight-line or double-declining depreciate scheme.

	Year 0	1	2	3	4	5	6
Capital investment	-150,000						
Rev - Exp		50,000	52,500	55,130	57,880	60,780	

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Rev - Exp		50,000	52,500	55,130	57,880	60,780	
SL Depreciation		30,000	30,000	30,000	30,000	30,000	
Tax (21%)		4,200	4,730	5,280	5,860	6,460	
OCF		45,800	47,780	49,850	52,030	54,310	

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Tax (21%)		4,200	4,730	5,280	5,860	6,460	
OCF		45,800	47,780	49,850	52,030	54,310	
DD Depreciation		60,000	36,000	21,600	16,200	16,200	
Tax (21%)		-2,100	3,465	7,042	8,753	9,360	
OCF		52,100	48,740	48,090	49,130	51,420	

Investment in Working Capital

Net working capital is the difference between current assets and current liabilities

- Increases in NWC typically occur at the beginning (increase in inventory and accounts receivables) = investment (cash outflow)
- Investment in NWC usually offset at end of project's life = payout (cash inflow)
- Make sure to remember the cash flow is the CHANGE in NWC